TrES-3b

R-0330

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_170419` · created 2026-07-18T04:35:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457862.82778 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.014
Transit depth (Rp/Rs)^2	3.82 +/- 0.56 [%]
Orbital Inclination (inc)	82.28 +/- 0.53
Airmass coefficient 1 (a1)	1.159 +/- 0.017
Airmass coefficient 2 (a2)	-0.1235 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	1.41 %
Best Comparison Star	None
Optimal Aperture	4.22
Optimal Annulus	8.29
Transit Duration (day)	0.0618 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.014 vs 0.16309 (deviation 1.66σ)
Duration fit vs published	0.0618 d vs 0.0567 d (deviation 0.85σ)
Mid-time O–C	2.5 min
Depth SNR	10.2
Residual scatter	1.41 %

Light curve

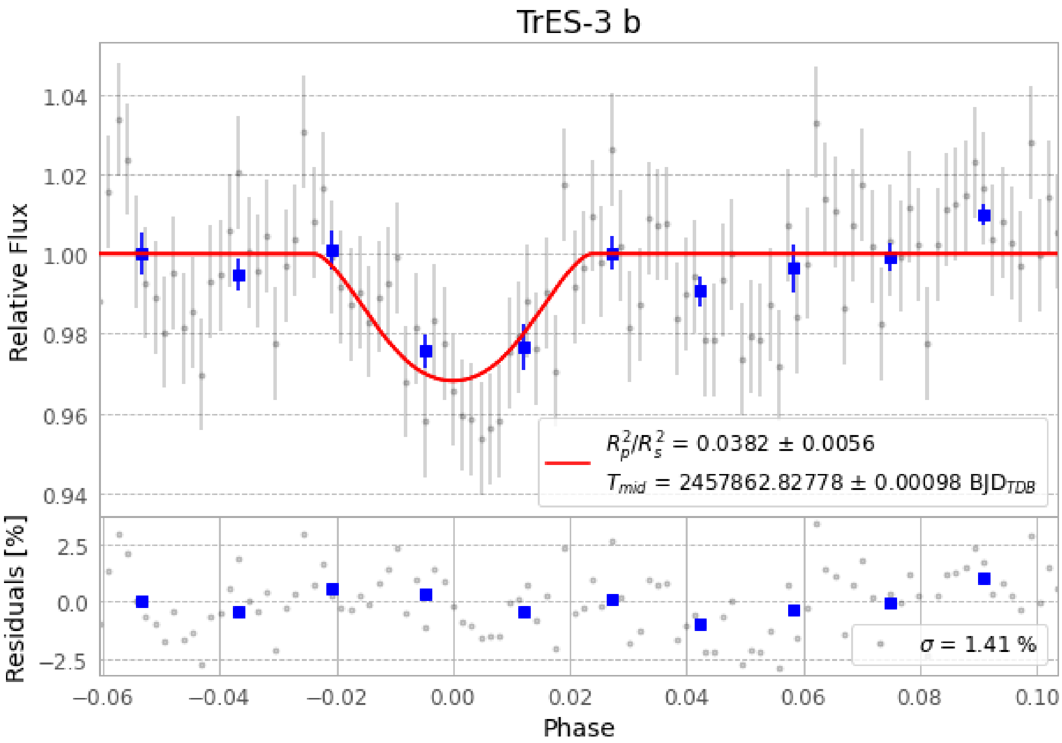

Figure 1 — FinalLightCurve_TrES-3 b_2017-04-18.png (SHA-256 99e612bd9318b69c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [180, 210], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 7c8f7a32edc77562…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit e9b0e33`

Data provenance

104 FITS frames (68 MB), `TRES-3170419055412.FITS` → `TRES-3170419110312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-170419.log (205a4630758e…), FinalLightCurve_TrES-3 b_2017-04-18.png (99e612bd9318…), FinalLightCurve_TrES-3 b_2017-04-18.csv (be87e4ce6f1a…)

Compute resources

Wall time 310.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-18 04:35Z.